

Hamdi Jarboui

Senior Python Engineer | AI Agents & LLM Automation | DevOps & CI/CD

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PROFESSIONAL SUMMARY

Senior Python engineer with 4+ years delivering production systems end to end, from architecture to deployment, across AI agents, DevOps and embedded automotive software. At KPIT I lead the design of AI agent automation and built the company's **AI harness**: 170 skills, 163 agents, 40 rules and 62 commands shared across four AI coding assistants, and a library of **639 MCP tools** across 29 servers that connects agents to CI, ALM, artifact and diagnostic systems. Hands-on with LLM integration (Claude, OpenAI), RAG, multi-agent orchestration and safe autonomy (human approval, risk-gated tools). Owner of CI/CD from commit to production, trainer of ~100 engineers in generative AI, and Python & applied-AI instructor at the ENETCOM engineering school.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash

AI Agents & LLMs: Agentic AI, multi-agent orchestration, Model Context Protocol (MCP) servers & clients, tool/function calling, RAG (LangChain, pgvector, OpenAI embeddings), knowledge graphs, prompt & context engineering, Anthropic Claude, OpenAI, Claude Code, OpenCode, GitHub Copilot

AI Safety: Human-in-the-loop approval, risk-tiered tool access, fail-closed merge gates, loop detection, secret redaction

Back end: Django, Starlette, REST APIs, asyncio, Pydantic, Node.js, Bun, Zod, React

DevOps & CI/CD: GitLab CI/CD, GitHub Actions, Jenkins, Zuul, Docker, Docker Compose, Caddy, Nexus, JFrog Artifactory, parallel CI jobs, tag-gated releases, private package registries, AWS, Linux

Testing & Quality: pytest, Robot Framework, contract testing, E2E testing, coverage gates, ruff, mypy, ESLint, TestGuide

Observability: Grafana, Prometheus, Kibana, Elastic, OpenTelemetry, Plotly Dash

Automotive: CAN/CAN-FD, DBC, UDS over DoIP, ODX/PDX, AUTOSAR ARXML, A2L, DLT, MDF4, Lauterbach TRACE32, MISRA C:2012, ISO 26262, ASPICE, ECU validation

Data & ML: PostgreSQL, SQLite, MySQL, MSSQL; machine learning, deep learning (TensorFlow), computer vision

Integrations: GitLab, GitHub, Jira, Confluence, Microsoft Graph / 365, Slack, Microsoft Teams

PROFESSIONAL EXPERIENCE

Senior Python Developer & DevOps Engineer, AI Agents

KPIT Technologies · Sfax, Tunisia · 07/2024 – Present

- Lead the design and delivery of AI agent automation in Python, setting the technical direction for how the team automates repetitive engineering work.
- Built **Anvil**, the company-wide AI harness: **170 skills, 163 agents, 40 rules and 62 commands** installed from one registry into Claude Code, OpenCode, Beacon and GitHub Copilot, with zero npm dependencies and tag-gated releases.
- Architected **ai-tools-library**: **29 MCP servers exposing 639 tools** for GitLab, Jenkins, Jira, Confluence, Grafana and automotive diagnostics, shipped through a **7-stage CI pipeline** with per-package coverage gates up to 98%.
- Built **MTF Assistant**, a RAG chatbot (Django, React, LangChain, pgvector) answering questions on the internal test framework, used by **~600 testers and developers**.
- Own CI/CD (Jenkins, Zuul, GitLab CI/CD) from commit to production, with Nexus/Artifactory artifact management, TestGuide test orchestration, Docker and AWS deployment.
- Work directly with customers to gather requirements and present technical solutions; delivered a generative AI workshop to the whole development department (~100 engineers).

Python Developer & DevOps Engineer

Primatec Engineering · Sfax, Tunisia · 07/2022 – 06/2024

- Reference engineer for Python ECU flashing and configuration tooling, enabling faster embedded-systems validation for automotive OEM programmes.
- Automated build, test and deployment pipelines with Jenkins and Zuul (including Zuul on AWS), removing manual steps from the release process.
- Built full-stack Python tools and dashboards adopted across engineering teams for data analysis and reporting.
- Trained developers and testers in ~20-person cohorts on Python testing (pytest, Robot Framework).

SELECTED PROJECTS

opencode-autodev: autonomous ticket-to-merge delivery with AI agents (solo, built at KPIT)

TypeScript · Bun · SQLite · OpenCode SDK · GitLab, GitHub, Jira, Microsoft To Do · 740 tests, 96% coverage

- Collects eligible tickets from GitLab, GitHub, Jira and Microsoft To Do, picks a specialist developer agent per ticket (e.g. Anvil agents) and implements each one in an isolated git worktree, with many tickets in parallel under SQLite leases.
- Developer agent, reviewer agent and the human operator share a per-ticket squad channel (@-mentions, questions mid-run) run by a deterministic facilitator with hard message and wake-up budgets.
- Merges only through a fail-closed four-way gate (reviewer verdict, **human approval**, pipeline, security jobs); config validation refuses auto-merge without at least one human approval. 15 ADRs, 85% coverage gate.

HCode: AI coding agent for the terminal (creator & core author, team project)

Python · Anthropic & OpenAI · MCP · 44k lines · 827 tests

- Autonomous coding agent built from scratch with a Plan-Execute-Verify workflow, automatic failover between Claude and OpenAI behind circuit breakers, three-layer memory (project, session, semantic) and per-task transactions.
- The same agent core powers a terminal UI, plus a desktop app and a VS Code extension built with two collaborators.

Automotive Test Case Generation Agents: from specification to traceable test suite

Python · LLM agents · RAG · knowledge graph · MCP · GitLab CI parallel jobs

- AI agents that turn automotive specifications into test cases, grounded through RAG over the requirement text and a knowledge graph linking each requirement to its signals, functions and dependencies.
- Agents call MCP tools for signal and interface definitions, related requirements and existing tests; generation fans out over parallel GitLab CI jobs, so a whole specification is covered in one pipeline run with every test traced to its requirement ID.

Full case studies with architecture and metrics: hamdijarboui.github.io/portfolio

TECHNICAL TRAINING & INSTRUCTION

5 years training engineers and students in Python, generative AI and agentic AI: 20-person cohorts, 10–30 hour programmes, **100% PCEP pass rate**, students placing 3 times at national AI competitions.

- **Lead Instructor**, Python, Generative AI & Computer Vision · 2S Training and Consulting · 2021 – Present
- **Python & Applied AI Instructor** · ENETCOM Engineering School, Sfax · 2024 – Present
- **Python & Generative AI Instructor** · ISB International School of Business · 2024 – Present
- **Python Instructor** · Smart Skills Academy · 2022 – 2023

EDUCATION

- **Engineering Degree, Industrial Computer Engineering** · National School of Electronics and Telecommunications (ENETCOM), Sfax · 2019 – 2024
- **Preparatory Cycle, Physics & Chemistry** · Faculty of Sciences, Sfax · 2017 – 2019

CERTIFICATIONS

- Python Institute: PCEP (Certified Entry-Level Python Programmer), PCAP (Certified Associate in Python Programming)
- IBM: Applied Data Science with Python (Level 2) · Deep Learning (Level 2), TensorFlow
- University of Washington (Coursera): Machine Learning Specialization
- Microsoft MTA: Programming Using Python & Java · Certiprof: Scrum Foundation (SFPC)

LANGUAGES

Arabic: native · French: full professional proficiency · English: professional working proficiency